

Deepak Kumar

Software Developer | Node.js specialist

☎ 7488404273 ✉ deepak.singh.tech48@gmail.com 📍 Bengaluru, Karnataka, India

Professional Summary

Skilled Software Developer with expertise in backend development and Generative AI integration, complemented by proficiency in full-stack web technologies. Engineered scalable, secure backend systems using Java, Spring Boot, and Microservices architecture, achieving up to 40% performance improvements. Mastered integration of Large Language Models (LLMs) using LangChain and OpenAI APIs to deliver intelligent applications.

Key Highlights:

- Focus on backend development using Spring Boot, Java, REST APIs, and Microservices
- Experience with Generative AI, including Prompt Engineering, LangChain, and LLM APIs
- Solid grounding in full-stack development using Next.js, React.js, and Node.js
- Skilled in cloud deployment and DevOps workflows using AWS, Docker, and CI/CD

Skills

- **Programming Languages:** C++, Python, Java, JavaScript (ES6+), TypeScript
- **Core CS Concepts:** OOP, DSA, Operating Systems, DBMS, Design Patterns
- **Frontend Development:** HTML5, CSS3, JavaScript, React.js, Next.js, Tailwind CSS, Bootstrap
- **Backend Development:** Node.js, Express.js, Django, Java (J2EE), Spring Boot, Spring MVC, Spring Security, RESTful APIs, Microservices Architecture
- **Generative AI & LLMs:** Prompt Engineering, LangChain, OpenAI API, Chatbot Dev, RAG, Semantic Search
- **Cloud & DevOps:** AWS, Docker, Kubernetes (basic), CI/CD, Git, GitHub, Netlify, Vercel, Jenkins (basic), GitLab CI
- **Databases:** MySQL, Oracle, MongoDB, NoSQL
- **Tools & Platforms:** VS Code, IntelliJ IDEA, Eclipse, Postman, Swagger, Figma (basic), JIRA
- **Soft Skills:** Logical Thinking, Problematic Solver, Team Collaboration, Communication, Task Management, Agile Scrum

Professional Experience

Inventeron Technologies and Business Solutions LLP

Jan 2025 – Jun 2025

Java Software Developer Intern

- Architected and deployed robust backend systems using Spring Boot, Java language and RESTful APIs, supporting enterprise-grade applications with 99.9% uptime.
- Optimized relational database schemas in MySQL, improving query performance by 30%.
- Implemented Spring Security to enforce secure authentication and role-based access, reducing unauthorized access risks by 100%.
- Designed and developed Microservices Architecture, enabling 25% faster scalability and maintainability for large-scale systems.

EISystems Services

May 2024 – Jul 2024

Data Science Intern

- Spearheaded a Play Store Rating Predictor using Python and Scikit-learn, achieving 85% prediction accuracy.
- Applied Linear Regression, Decision Trees, Random Forest, and Gradient Boosting models, optimizing performance with 20% lower RMSE.

ITJOBXS

Jun 2023 – Aug 2023

Software Engineer Intern – LMS Portal Developer

- Engineered an interactive Learning Management System (LMS) using ReactJS, Styled Components, and Hooks, increasing user engagement by 35%.
- Designed responsive user interfaces using Jade, Stylus, and BackboneJS, improving cross-device compatibility by 40%.
- Streamlined development processes through CI/CD pipelines and Agile methodologies, reducing deployment time by 15% in a remote team setting.

Projects

Study Notion *Node.js, Express.js, MongoDB, JWT*

- Engineered a full-stack ed-tech platform with secure login, course creation, and live sessions, supporting over 5,000 active users.
- Integrated JWT and Bcrypt for secure authentication, reducing security vulnerabilities by 100%.

AI Mock Interview System *Next.js, Tailwind CSS, Vapi, Firebase, Gemini API*

- Implemented an AI-powered mock interview platform using voice AI agents and Google Gemini API, improving user preparedness by 40%.
- Drafted interactive UIs and integrated session tracking endpoints, increasing user interaction rates by 30%.

Realtime Device Tracker *Node.js, Express.js, Socket.io, Leaflet.js*

- Built a real-time device tracking system using WebSockets and Leaflet.js, achieving 95% accuracy in map visualizations.
- Developed backend services to update and fetch device coordinates, reducing latency by 20% through enhanced socket events.

Research

Image Compression Model Using Deep Learning *NIT Patna / Prof. Amit Kumar Singh*

- Advanced a CNN-based image compression model optimizing quality/size by 30%.
- Established a comprehensive benchmarking protocol by employing PSNR and SSIM metrics, leading to a 25% reduction in data storage costs while ensuring optimal video quality for seamless customer experiences.

Education

NIT Patna

Bachelor of Technology in Computer Science and Engineering

S.N. Sinha College, Aurangabad

Intermediate (10+2), Science Stream

S.S.M School, Aurangabad

Matriculation (10th)

Positions of Responsibility

Core Member, GDSC – NIT Patna

Core Team, Hackofest – Annual Hackathon @NIT Patna

Member, Think India – NIT Patna Chapter

Core Member, Coding Club – NIT Patna

Achievements

Solved 1100+ DSA problems (LeetCode, GFG, CodeStudio)

LeetCode Rating: 1704 CodeChef: 1600

Technical Lead, Hack-o-Fest, NIT Patna – National level hackathon

JEE Mains 2021: 98.14 percentile (Top 0.92%)

Certifications

- DSA – Coding Ningas
- Web Development – CodeHelp
- DBMS – Coding Ningas
- Machine Learning – Coding Ningas
- AI & ChatGPT Essentials – Coding Ningas
- Generative AI – IIT Guwahati
- Backend Engineering – Airtribe